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How Can *Bitcoin* Be Worth Anything?

A first-principles essay on value, scarcity, and money —
from the paradox of water and diamonds to the twenty-
one million coins that cannot be printed.

EDITED BY

Ratty2Tails

ADAPTED FROM A RECORDED LECTURE, "BITCOIN 101"

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How Can Bitcoin Be Worth Anything?

An honest, first-principles exploration of value — and how it applies to bitcoin

Edited by **Ratty2Tails** · Monetary History Series · The Ratty2Tails Papers

TL;DR

Value does not live inside a thing. It lives in the relationship between a good and a human need — a judgment made in a person's head, not a formula baked into the object. Economics exists at all only because humans want more than they can have; that gap forces choice, and choice is where subjective value is made.

A good becomes economic the moment it is both useful and scarce — meaning demand for it exceeds the available supply. Air is useful but not scarce, so it is free. A one-of-a-kind rock is scarce but not useful, so it is worthless. Bread, lakehouses, gold, and bitcoin are both, so we must economize around them.

Prices are not set by governments, companies, or committees. They are the residue left over after millions of people compare their options and choose, one marginal decision at a time — which is why water, essential to life, is cheap, while diamonds, essential to nothing, are expensive.

Bitcoin has no cash flow, no company, and no government guarantee behind it. But it is useful — fixed and verifiable rules, self-custody, portability, censorship resistance — and it is scarce, with a hard cap no institution can inflate away. Usefulness creates the demand; scarcity makes that demand consequential. Together, the two produce a price.

Introduction and Scope

The skeptic's question is a fair one, and Bitcoiners do themselves no favors by waving it off. Bitcoin has no cash flow. It has no company behind it, no government guarantee, no redemption for metal in a vault. Stripped to its mechanics, it is information stored on computers. So why is that information worth anything at all?

This essay does not start with Bitcoin. It starts with the older and harder question underneath it: why is anything worth anything? Before we can ask whether bitcoin

deserves a price, we need a working theory of where prices come from in the first place — for water, for oil, for a lakehouse, for a dollar bill. Once that foundation is in place, bitcoin's value stops looking mysterious and starts looking like an ordinary case of an old and well-understood pattern.

What follows is an edited essay adaptation of a recorded lecture — the first installment of a "Bitcoin 101" series — reworked here into essay form for the Monetary History Series. The argument and examples are the original speaker's; the structure, prose, and framing are the editor's.

I. Value Is Not Inside the Thing

Cut a Ferrari open and you will not find, tucked somewhere behind the seat cushion, the formula that makes it worth more than a Honda. Cut open a designer handbag, a house, a diamond — the value is not in there. This is the first and most disorienting fact about economics: value is not a property of objects. It is a relationship between an object — what economists call a good — and a human need.

That relationship is entirely subjective, resting on how much a person wants something and how well they believe the good will satisfy that want. And because human wants shift constantly — sometimes a person urgently needs a thing, sometimes they have plenty and need no more — the value assigned to any given good is never fixed. It moves with the person, not with the object.

Rarity alone proves nothing. A child's kindergarten painting may be one of a kind; that does not make it valuable. A uniquely-shaped rock found in a particular backyard is scarce by definition, but listing it for sale guarantees nothing, because scarcity without a satisfied need is worthless. Value, at bottom, is about satisfying a human need. Where there is no need, there is no value — no matter how rare the thing is.

II. Why Economics Exists at All

Economics would not need to exist if we could have everything we wanted. That single sentence is the entire justification for the discipline. There is a permanent gap between wanting something and having it, and that gap is what forces every human being to make choices.

Humans are a future-oriented species. We imagine better lives for ourselves — a degree, a business, a new city, a relationship — and we cannot pursue all of those futures simultaneously. One life, several branching paths, and no way to walk down all of them at once. This is not yet an economic decision in the narrow sense; it is simply the human condition. Scarcity, in this light, is not a shortage to be pitied — it is the basic fact that choosing one path means declining the others. For every yes, there is a no.

ON ABUNDANCE

Automation and AI may well deliver an abundance of goods undreamt of by kings a few centuries ago — and to that extent, optimists like Elon Musk are right that material scarcity is receding on many fronts. But abundance in some categories does not abolish the human condition. We will always want more than any given moment allows: a better car, a longer life, a bigger house, more time. As long as wanting outpaces having, the need to choose — and therefore to economize — persists.

It follows that scarcity, correctly understood, simply means the necessity to choose. It does not, by itself, imply suffering. It implies that options A through F cannot all be taken, and someone has to decide which one is most valuable to them, in their own circumstances, right now.

III. What Makes a Good "Economic"

Not everything we value requires economizing. Air is essential — without it, we die within minutes — yet nobody pays for it, because supply so vastly exceeds demand that no exchange is necessary. Air is useful, but it is not scarce, so it is not an economic good.

Flip the case: a one-of-a-kind rock found in a particular yard is technically scarce — there is exactly one of it — but it is useless. Picking it up costs nothing and matters to no one. It, too, fails to become an economic good, for the opposite reason.



Figure 1. A good becomes economic only where usefulness and scarcity overlap.

Scarcity alone (bottom right) is not enough; usefulness alone (top left) is not enough.

A lakehouse sits in the useful-and-scarce quadrant: there is only so much lakefront land, and a lakehouse can genuinely shelter, sleep, and host people. So does a cheeseburger, in a smaller way — not scarce in the sense of being finite like gold, but scarce in the sense that more people want to eat than there is food available at any given moment. "Scarce," in this technical sense, never means merely rare. It means more demand than supply.

Value assigned to a good, moreover, is not permanent, because usefulness itself can change. In the 1800s, crude oil was industrial sludge — a nuisance landowners wanted off their property. The molecules had not changed by the time the combustion engine arrived and oil became the fuel underwriting the entire global economy; what changed was how well oil could satisfy a human need. The same physical substance moved from worthless to indispensable purely because the relationship between the good and human wants shifted.

IV. The Water–Diamond Paradox

This brings us to one of economics' oldest puzzles: why is water — without which we die in days — so much cheaper than diamonds, which nobody needs to survive? The

instinctive answer, that water is simply more "abundant," is true, but it hides the more interesting mechanism underneath.

Nobody, when checking out at a grocery store, is weighing all the water in the world against all the diamonds in the world. They are weighing one more bottle of water against the other things they could do with the same few dollars. That is the crucial insight: prices emerge at the margin — from the next unit, not from the category as a whole.

We value marginal units. We do not value abstract totals.

IN THE KITCHEN	DYING OF THIRST IN THE DESERT
A bottle of water costs roughly a dollar, trending toward zero with a home filtration system.	The same bottle of water — same H ₂ O, nothing physically different — could command a person's entire net worth.
The next bottle satisfies a minor, easily deferred want.	The next bottle satisfies the most urgent want a person can have: survival.
Circumstances: abundant access, low urgency.	Circumstances: scarce access, maximum urgency.

Nothing about the water changed between these two scenes. What changed was the person's circumstances, need, and available alternatives — and that is precisely what subjective valuation means. Water, taken as a category, is more valuable to humanity than diamonds by any sane measure; deprived entirely of one or the other, civilization would only survive the loss of diamonds. But nobody trades at the level of "humanity as a whole." Every trade happens at the margin, between one buyer, one seller, and the next unit of the good in question — which is why an object essential to life can carry a lower price than an object essential to nothing.

V. Every Trade Has Two Winners

A point that even seasoned observers of markets routinely miss: in a free exchange, both sides win, by their own subjective reckoning. When a buyer hands over two dollars for a bottle of water, the purchase itself is a statement — the water is worth more to that buyer than the two dollars, or the trade would not have happened. The seller is making the mirror-image claim: the two dollars is worth more to them than the water, perhaps because they can restock at wholesale prices far below retail.

Neither party needs to know, or agree with, the other's reasoning. The buyer has no idea whether the seller is desperate for cash or simply optimizing margins; the seller has no idea how thirsty the buyer is. Each side is only ever pricing the trade against their own situation — and because both walk away believing they gained, free exchange is not a zero-sum contest but a mutual, if asymmetric, creation of value. This logic holds without exception in voluntary trade: coerced exchange — a trade backed by threats rather than consent — is a different phenomenon entirely, and not the one under discussion here.

The same logic applies to someone selling bitcoin to cover a medical bill, or buying bitcoin to escape a currency in freefall. Both are, in that moment, valuing what they receive more than what they give up. Neither is wrong; each is simply comparing two options against their own circumstances, which is the entire content of a "free market."

VI. The Three Jobs of Money

Before asking what makes bitcoin valuable specifically, it helps to be precise about what money is for. Money performs three distinct jobs.

It moves value across time. Work performed today does not have to be consumed today. Money lets a person hold the value they created until they are ready to spend it — next week, next year, or in retirement decades away.

It moves value across space. Value created in one city needs a way to reach another — a supplier in a different country, a family member overseas, a purchase made on the other side of the planet.

It moves value across choices. This third job is the one most relevant here. Money lets radically unlike things be compared. A software engineer has no idea what it takes to raise cattle; a butcher has no idea what elliptic curve cryptography is. Money is the common unit that lets each of them price their output against the other's, so that a fair exchange can happen despite neither side understanding the other's trade.

Every candidate for "good money" — dollars, gold, pounds, bitcoin — is ultimately being evaluated against these three jobs, and against a short but consequential list of things that can go wrong: the money can be inflated by someone else's decision to print more of it; it can be seized if held by a third party; it can be expensive to move through layers of intermediaries; its rules can be changed without the holder's consent; and it can be impossible for the holder to verify independently.

VII. Why Bitcoin Has Value

With that foundation, bitcoin stops requiring a special exception to economic theory. It requires only the same two questions asked of water, oil, and lakehouses: is it useful, and is it scarce?

On usefulness: bitcoin can be held by its owner directly, at effectively no cost, without a vault, an auditor, or a custodian standing between the holder and their savings. It can move anywhere on earth for a small fee. Nobody can unilaterally issue more of it. Anyone can verify, with mathematics rather than trust, that the rules have not quietly changed overnight. For a person who wants exactly those properties — fixed supply, self-custody, portability, and rules nobody can alter at will — bitcoin does something no prior form of money did as completely.

Crucially, nobody has to authorize that want. No government, company, or platform approves a person's decision to find bitcoin useful; they simply do, or they do not, based on their own circumstances. The early history of Bitcoin makes this vivid. The Silk Road marketplace, whatever one thinks of what was traded there, demonstrated real demand for a currency that could move value for a transaction no existing payment rail — not cash, not PayPal, not a bank wire — was willing or able to service. That demand was not manufactured by an institution; it emerged because bitcoin was useful to a specific set of people with a specific need, before it was liquid enough to be called "digital gold" or listed in an ETF.

The same logic extends to a person living under a currency collapsing from hyperinflation, or a person fleeing an authoritarian regime who needs to carry meaningful wealth across a border in a form that cannot be confiscated at a checkpoint the way gold or real estate can. Nobody has to find that use case compelling personally for it to generate real demand from someone who does.

On scarcity: bitcoin's supply is capped at twenty-one million, enforced not by a promise but by rules any participant can independently check. If demand for bitcoin rises, the response cannot be "print more" — the way a government can respond to rising demand for its currency, or a company can issue more shares. Supply is fixed regardless of demand, so any increase in demand can only be met by a higher price, not a larger supply. That is the scarcity half of the equation, and it only matters because the usefulness half is already present.

It is important not to reverse the causality here. Bitcoin is not valuable because it is scarce — a scarce, useless object (the one-of-a-kind backyard rock) proves that scarcity alone creates nothing. Bitcoin is valuable because it is useful to real people solving real problems, and its scarcity is what prevents that demand from being diluted away by unlimited new supply. Scarcity does not create the demand; it makes the demand consequential.

One common objection deserves a direct answer: since bitcoin is open-source, can't anyone simply copy the code and create an identical asset? Technically, yes — but copying the code does not copy the network. A forked coin inherits none of the existing holders, none of the accumulated proof-of-work securing the ledger, none of the liquidity, and none of the social consensus that makes the original bitcoin useful for the jobs described above. It is the difference between copying Twitter's source code and copying Twitter's actual users: the software is trivial to duplicate, and worthless to duplicate, because the value was never in the code alone.

VIII. Choices at the Margin

Bitcoin, like every other candidate for savings, is competing against cash, stocks, bonds, gold, and real estate for a share of what people decide to do with money left over after their bills are paid. Nobody sits down and asks, in the abstract, "is bitcoin objectively more valuable than water?" — any more than a shopper asks whether diamonds are objectively more valuable than water while standing in a jewelry store. The only question that actually gets asked, over and over, by millions of separate

people, is: given what I have left over, and given my own view of the alternatives, where does the next dollar go?



No committee, company, or government sits at any point in this chain.

Figure 2. Price is not set — it is what remains after every participant in a market has separately compared their own alternatives and chosen.

Every current bitcoin holder, every day, is implicitly re-making the decision to hold rather than sell — into cash, real estate, equities, or anything else on offer. Every prospective buyer is asking the mirror question: would I rather hold this than the alternative in front of me? Multiply that decision across everyone participating in the market, at every price, and what is left over — the point where willing buyers and willing sellers actually meet — is the price. Nobody sets it. Everybody makes it, together, without coordinating.

SUMMARY

Scarcity does not make bitcoin valuable on its own. Bitcoin's usefulness creates demand for it, and its scarcity makes that demand matter — and the interaction of the two, worked out across millions of individual, marginal choices, is what results in a price.

This is not a special exception carved out for bitcoin. It is the same rule that governs water, oil, lakehouses, and every other economic good: something has value the moment someone, for their own reasons, judges it useful enough to economize for — and it keeps that value only for as long as it remains both useful and scarce.

Editor's Note

This essay is an edited prose adaptation of the first lesson in a recorded "Bitcoin 101" series, transcribed from spoken lecture into essay form for the Monetary History Series of The Ratty2Tails Papers. The argument, structure of reasoning, and illustrative examples — the lakehouse, the desert bottle of water, the Silk Road, the water-diamond paradox — belong to the original speaker. The prose, section structure, figures, and framing in this edition are the editor's own work, undertaken to render a spoken lecture into a form suited to careful, repeated reading.

Where the original lecture used first-person asides and conversational digressions, this edition renders the same points in expository prose, trims repetition native to spoken delivery, and adds two original diagrams (Figures 1 and 2) to make the underlying structure of the argument — usefulness, scarcity, and the marginal origin of price — visible at a glance.

Source & Further Reading

1. Original source recording: "Lesson 1: How Can Bitcoin Be Worth Anything?" — Bitcoin 101 lecture series, YouTube, <https://www.youtube.com/watch?v=ad3EvYEpuOQ>
2. Menger, Carl. *Principles of Economics* (1871) — the foundational statement of subjective, marginal value theory underlying this essay's argument.
3. von Mises, Ludwig. *Human Action: A Treatise on Economics* (1949) — praxeology, scarcity, and the theory of choice.
4. Menger, Carl. *On the Origin of Money* (1892) — the classical account of how a good emerges as money through marketable liquidity, relevant to bitcoin's monetization.
5. Nakamoto, Satoshi. *Bitcoin: A Peer-to-Peer Electronic Cash System* (2008) — the original protocol design referenced throughout as the source of bitcoin's fixed and verifiable supply rules.